1.4 States of Matter

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.4 States of Matter
Difficulty	Easy

Time allowed: 10

Score: /8

Percentage: /100

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Question 1

Which solid has a giant covalent lattice structure?

A. Calcium fluoride

B. Nickel

C. Silicon (IV) dioxide

D. Sulfur

[1 mark]

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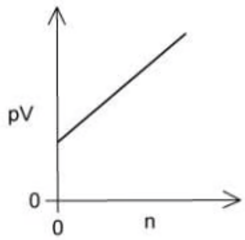
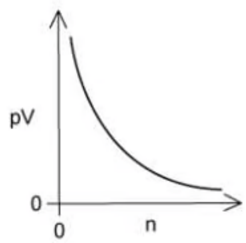
Question 2

For an ideal gas at constant pressure and temperature, which diagram shows the correct graph of pV against n?

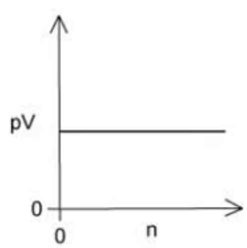
A.



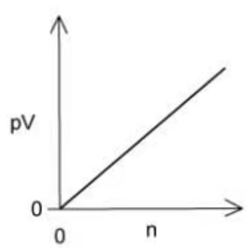
B.



C.



D.



[1 mark]

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Question 3

The table shows the physical properties of four substances.

Which substance has a giant ionic structure?

	melting point /℃	boiling point /℃	electrical conductivity of solid	electrical conductivity of liquid	electrical conductivity of aqueous solution
1	-119	39	poor	poor	insoluble
2	-115	-85	poor	poor	good
3	993	1695	poor	good	good
4	1610	2230	poor	poor	insoluble

A. 1, 2 and 3

B. 1 and 2

C. 1 and 3

D. 3 only

[1 mark]

Question 4

The M_r value of a gas can be calculated from the general gas equation.

Which expression will give the value of M_r for a sample of a gas of mass *m* grams?

A. $M_r = \frac{mRT}{pV}$

B. $M_r = \frac{pVRT}{m}$

C. $M_r = \frac{mpV}{RT}$

D. $M_r = \frac{pV}{mRT}$

[1 mark]

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Question 5

Which of the following least resembles an ideal gas at room temperature?

A. Helium

B. Ammonia

C. Carbon dioxide

D. Hydrogen

[1 mark]

Question 6

Three substances, R, S and T, have physical properties as shown.

Substance	Melting / °C	Boiling / °C	Electrical conductivity of liquid
R	772	1407	Yes
S	114	183	No
T	1610	2205	No

What type of lattice could each substance have?

	R	S	T
A	Giant molecular	Simple molecular	Ionic
B	Ionic	Giant molecular	Simple molecular
C	Ionic	Simple molecular	Giant molecular
D	Simple molecular	Ionic	Giant molecular

[1 mark]

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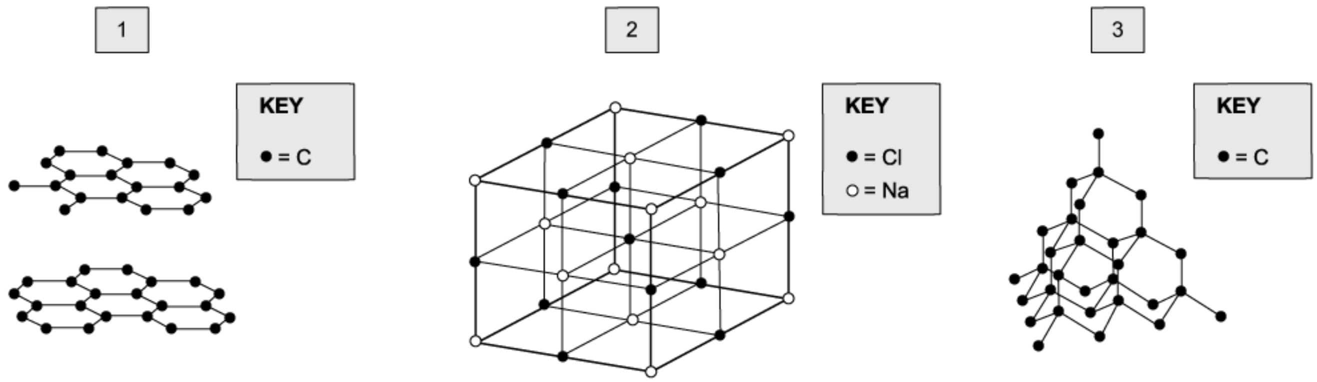
Question 7

Which diagram represents part of a giant covalent lattice structure?



A. 1, 2 and 3

B. 1 and 2

C. 1 and 3

D. 3 only

[1 mark]

Question 8

The ideal gas equation below summarises the gas laws.

$pV = nRT$

Which statement below is correct?

A. There exist intermolecular forces of attraction between gas molecules.

B. Ideal gas molecules will collide inelastically upon impact with each other.

C. One mole of an ideal gas occupies the same volume under the same conditions of temperature and pressure.

D. The volume of a given mass of an ideal gas is doubled if its temperature is raised from 25 ℃ to 50 ℃.

[1 mark]

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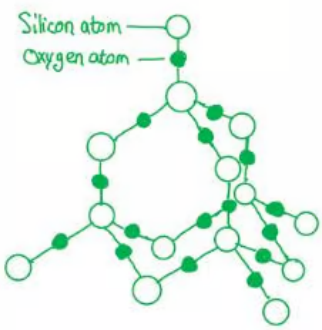
Model Answers: Easy

1

The correct answer is **C** because:

- Silicon dioxide or silica is the main component of sand.
- It is an example of a substance with a giant covalent structure.
- Silicon and oxygen are non-metals bonded together with covalent bonds.
- The atoms are joined together in a regular arrangement forming a lattice structure.
- Each silicon atom is bonded to 4 oxygen atoms, but each oxygen atom is bonded to only 2 silicon atoms giving the formula SiO_2.



A is incorrect as calcium fluoride is an ionic compound.

B is incorrect as nickel is a metal with metallic bonding.

C is incorrect as elemental sulfur is found as a cyclic S_8 molecule.

2

The correct answer is **D** because:

- The ideal gas equation is ***pV = nRT***

  - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant (8.314 J K^{-1} mol^{-1}), and T is the temperature (K).

- **Avogadro's law**: For a confined gas, the volume (V) and the number of moles (n) are directly proportional if the pressure and temperature both remain constant.
- The graph that shows a directly proportional relationship; that is a straight line and goes through the origin, so graph **D.**

A is incorrect as this line does not go through the origin so does not show a directly proportional relationship.

B is incorrect as this is not a straight line so does not show a proportional relationship.

C is incorrect as this graph shows as volume increase pressure stays constant.

3

The correct answer is **D** because:

- Ionic compounds have the following properties:
  - Does not conduct electricity as a solid.
  - Will conduct electricity as a liquid or molten.
  - High melting and boiling points.
- The only option that fits the properties above is substance 3.

4

The correct answer is **A** because:

- The molar mass of a gas can be derived from the ideal gas law, ***pV = nRT.***
- The definition of molar mass can be used to replace n in the equation.
- The M_r of a substance is defined as the mass of a substance occupied by 6.022 × 10^{23} molecules, this is also known as **Avogadro's number** and is the equivalent of 1 mole, therefore:
  - $moles\ (n) = \dfrac{mass\ (m)}{molar\ mass\ (M_r)}$

- **This can be substituted for *n* in the ideal gas law equation:**
  - $pV = \frac{m}{M_r} \times RT \quad \Rightarrow \quad M_r = \frac{mRT}{pV}$

5

The correct answer is **B** because:

- Ideal behavior is identical particles with no internal structure, no volume, random motion, no intermolecular forces and no loss of momentum in collisions.
- Ammonia has a strong force of intermolecular interactions (hydrogen bonding).

A is incorrect as helium is the one that will behave most like an ideal gas at room temperature due to the weaker intermolecular forces on the monoatomic molecule.

C is incorrect as carbon dioxide will only experience dispersion forces which are weaker than hydrogen bonds.

D is incorrect as hydrogen will only experience dispersion forces which are weaker than hydrogen bonds.

6

The correct answer is **C** because:

- Ionic compounds have the following properties:
  - Do not conduct electricity as a solid.
  - Will conduct electricity as a liquid or molten.
  - High melting and boiling points.
  - Substance R is an ionic lattice.
- Simple molecular compounds have the following properties:
  - No chemical bonding between the molecules, just dipole-dipole forces.
  - Low melting and boiling points.
  - Substance S has a simple molecular lattice structure.
- Giant molecular lattices have the following properties:
  - Three-dimensional arrangement of atoms.
  - Solid at room temperature.

- Very high melting and boiling points.
- Electrical insulators.
- Substance T is a giant molecular lattice structure.

7

- Giant molecular lattices have the following properties:
  - Three-dimensional arrangement of atoms.
  - Solid at room temperature.
  - Very high melting and boiling points.
  - Electrical insulators.
  - Substance T is a giant molecular lattice structure.

7

The correct answer is **C** because:

- The structure in option 1 is graphite.
- Graphite has strong covalent bonds with 3 other carbon atoms; this forms a giant layered structure with delocalised electrons.
- The structure represented by the diagram in option 3 is diamond.
- Diamond has strong covalent bonds with 4 other carbon atoms; this forms a regular tetrahedral structure with no free electrons.
  - Diamond has a giant covalent lattice structure.

Diagram 2 shows the structure of sodium chloride, a giant ionic structure.

8

The correct answer is **C** because:

- The ideal gas equation is ***pV = nRT***
  - Where *p* is pressure (Pa), *V* is the volume (m^3), *n* is the number of moles, *R* is the gas constant (8.314 J K^{-1} mol^{-1}), and *T* is the temperature (K).
- **Avogadro's law**: For a confined gas, the volume (*V*) and the number of moles (*n*) are directly proportional if the pressure and temperature both remain constant.

A is incorrect as the presence of intermolecular forces is not a gas law. An ideal gas will have no intermolecular forces between the molecules.

B is incorrect as in an ideal gas particles need to collide elastically (i.e. not lose momentum).

D is incorrect as the temperature is in °C, not kelvin. **Charles's law** states that the volume of a given amount of gas is directly proportional to its temperature on the kelvin scale when the pressure is held constant.